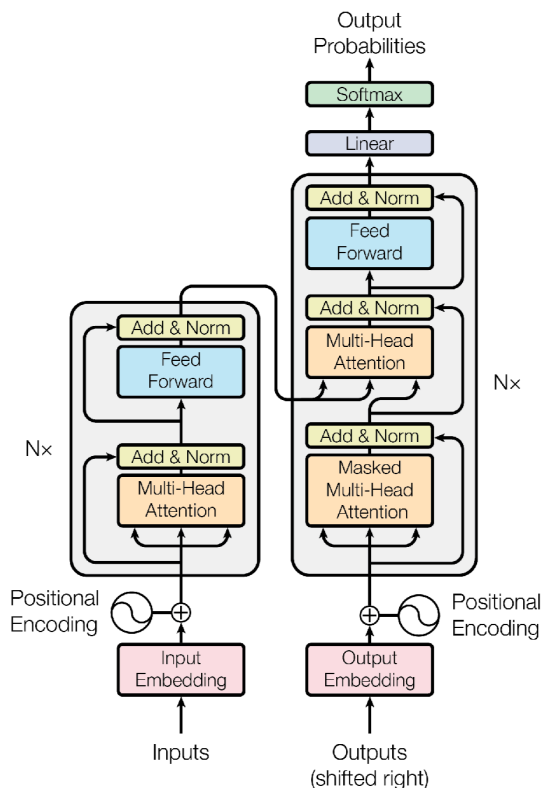


Transformers

Transformers, introduced in the paper "Attention is All You Need" by Vaswani et al. in 2017, revolutionized deep learning model architecture. Developed to address sequence transduction problems, such as neural machine translation, transformers handle tasks transforming input sequences to output sequences (e.g., speech recognition, text-to-speech transformation).



A key innovation within transformers is self-attention, enabling the model to weigh input positions differently during predictions. The encoder produces a representation equal in length to the original input sequence. During decoding, the context vector is dynamically generated through a weighted sum of input representations at each time step. Differentiating these weights is crucial, allowing the model to learn attention assignment alongside other neural network parameters.

In transformers, self-attention operates in a multi-head fashion, performing attention multiple times simultaneously. Each attention head scrutinizes different facets, constructing unique value vectors for a comprehensive understanding. As transformers lack inherent understanding of token sequence order, positional encoding is added to input embeddings to provide position information.

The positional encoding function typically combines sine and cosine functions with different frequencies, allowing the model to capture various positional patterns and avoid aliasing effects.

Transformers have become a versatile and powerful tool across domains like NLP, computer vision, speech recognition, recommendation systems, and text generation. Their ability to handle intricate patterns and relationships in diverse data types underscores their significance in modern deep learning applications.

